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For our project we made a Bluetooth gaming controller that uses the ESP32-WROOM-32D microcontroller chip because it had built-in bluetooth and was able to connect to most devices used for gaming, also because you can get 3 breakout boards for less than 30 dollars. After looking through other references and examples we were able to get our pcb board to have the same 14 buttons as a standard controller 4 for the Dpad, 4 for the action buttons ABXY, and 4 for the 2 pairs of trigger buttons on the top of a controller. As we kept following along the design process we were able to see how we could transform our initial block diagram to our final P.O.C prototype. We did halt some ideas like adding a ERM haptic motor for feedback vibrations and we still have a wireless version in the works but we did this as a way to focus on the core functions of the controller. We were also having issues putting the analog sticks in the schematic and pcb and settled on showing them as two potentiometers for the x and y axis for each joystick because a lot of modern gaming controllers use potentiometers to determine joystick placement. We made a voltage regulator that should give a constant 3.3 volts to power the microcontroller and other components that need power. Our PCB will have 4 layers to work with with the back having the voltage regulator circuit and the front with the microcontroller, analog sticks, and buttons. The middle layers will be for ground and the 3.3 volts so it will be easier to create paths to the other components. We used vias to create paths for ground and power but also as a way to make clean paths for our components, we are working on making the path ways more organized in future developments.

We also added leds to show that power is on and that the bluetooth is connected to a device to help any users know if the controller is on and if it is connected to a device. Overall we came a far way from where we started, going from drawings and ideas asking can we even do this or is this possible? To conversing on how much more we can do and what else we can add to our PCB to make it better.

We got far despite the challenges we faced and learned a lot in the process and hope there is a chance others can learn from what we made.